

ANALYTICAL GEOMETRY SOLUTION & VERIFICATION

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1. Problem Description & Coordinate Setup

The provided Python script simulates a geometric configuration inside and outside a square $ABCD$. By examining the coordinate definitions in the source code, we reconstruct the exact geometric parameters of the system:

- Let $ABCD$ be a square with an arbitrary side length $s = AB$. The vertices are placed on a Cartesian plane as:
 $A = (0, s)$, $B = (0, 0)$, $C = (s, 0)$, $D = (s, s)$.
- A point F lies on the top edge AD , located at a distance FD to the left of vertex D . Thus, its coordinates are $F = (s - FD, s)$, making the segment length $|DF| = FD$.
- An external point E is generated relative to vertex D at a specified distance and angle. According to the script, the segment length $|DE| = FD$.
- The position of E is given by:

$$\begin{aligned}x_E &= x_D + FD \times \cos(20^\circ) \\ y_E &= y_D + FD \times \sin(20^\circ)\end{aligned}$$

This means that the vector DE forms an angle of 20° counter-clockwise relative to the positive x-axis.

2. Purely Analytical Proof

To find the exact value of the angle $\angle AFE$ analytically, we look at the triangle $\triangle FDE$ formed by the vertices F , D , and E .

Step 1: Determine the properties of $\triangle FDE$.

By definition from the simulation configuration, the lengths of the two sides meeting at vertex D are equal:

$$|DF| = |DE| = FD$$

Because two sides are congruent, $\triangle FDE$ is an **isosceles triangle** with its vertex angle at D .

Step 2: Compute the interior vertex angle $\angle FDE$.

The direction from D to F is straight to the left along the segment DA , which corresponds to an angular direction of 180° . The direction from D to E is given as 20° from the horizontal. Therefore, the interior angle $\angle FDE$ is the angular difference between these two rays:

$$\angle FDE = 180^\circ - 20^\circ = 160^\circ$$

Note on the script's reference to **110°**: The downward vector **DC** points at **270°**. The angle between the downward vector **DC** (**270°**) and the vector **DE** (**20°**) is $270^\circ - 20^\circ = 250^\circ$ or externally $360^\circ - 250^\circ = 110^\circ$, which matches the script's visual annotation for $\angle CDE = 110^\circ$.

Step 3: Compute the base angles of $\triangle FDE$.

Since $\triangle FDE$ is isosceles with $|DF| = |DE|$, its base angles $\angle DFE$ and $\angle DEF$ must be equal. The sum of interior angles in any triangle is **180°**:

$$\angle DFE + \angle DEF + \angle FDE = 180^\circ$$

Substituting $\angle DFE = \angle DEF$ and $\angle FDE = 160^\circ$:

$$2 \times \angle DFE + 160^\circ = 180^\circ$$

$$2 \times \angle DFE = 20^\circ$$

$$\angle DFE = 10^\circ$$

Step 4: Compute the target angle $\angle AFE$.

The points **A**, **F**, and **D** lie on a straight line segment forming a straight angle of **180°**. The angle $\angle AFE$ and the interior triangle angle $\angle DFE$ are supplementary:

$$\angle AFE = 180^\circ - \angle DFE$$

$$\angle AFE = 180^\circ - 10^\circ = 170^\circ$$

Conclusion:

For any random positive choice of square side length **AB** and segment length **FD**, the value of the target angle remains perfectly invariant:

$$\angle AFE = 170^\circ$$

3. LaTeX Formatting

The standard LaTeX notation for this analytical proof is provided below:

```
\documentclass{article}
\usepackage{amsmath}
\begin{document}
\section*{Analytical Proof of Angle  $\angle AFE$ }
Given an isosceles triangle  $\triangle FDE$  where  $DF = DE = FD$ .
The interior vertex angle is computed as:
\begin{equation}
\angle FDE = 180^\circ - 20^\circ = 160^\circ
\end{equation}
Since  $\triangle FDE$  is isosceles, the base angles are:
```

```

\begin{equation}
\angle DFE = \frac{180^\circ - \angle FDE}{2} = \frac{180^\circ - 160^\circ}{2} =
10^\circ
\end{equation}
Since points $A, F, D$ form a straight line, $\angle AFE$ is supplementary to $\angle DFE$:
\begin{equation}
\angle AFE = 180^\circ - \angle DFE = 180^\circ - 10^\circ = 170^\circ
\end{equation}
\end{document}

```

References:

MIT Mathematics Explorer Query Reference: [usa_2021_c9c535](https://mathnet.mit.edu/explorer.html?p=usa_2021_c9c535) (https://mathnet.mit.edu/explorer.html?p=usa_2021_c9c535)